

MA388 Sabermetrics: Lesson 27

Launch Angles and Exit Velocity

```
library(tidyverse)
library(knitr)
library(baseballr)
```

After seeing a rise in home runs starting in 2015, Baseball Commissioner Robert D. Manfred, Jr. assembled a commission to analyze the situation. A press release outlining the commission's findings and MLB's actions can be found [here](#).

Let's take a look at 2017 Statcast data.

```
# # Statcast season window for 2015 (regular + early postseason window as example)
# start_date <- as.Date("2015-04-02")
# end_date   <- as.Date("2015-11-01")
#
# # Break the range into chunks (API is happier with day- or week-sized chunks)
# date_chunks <- seq(start_date, end_date, by = "1 week")
# date_ends   <- pmin(date_chunks + 6, end_date)
#
# # Pull BIP for each chunk
# statcast_2015 <- map2_dfr(date_chunks, date_ends, ~{
#   message("Fetching: ", .x, " to ", .y)
#   Sys.sleep(0.75) # be polite to the endpoint
#   statcast_search(start_date = .x, end_date = .y)
# })
#
# write_csv(statcast_2015, 'statcast_2015.csv')
#
# bip_2015 <-
#   statcast_2015 |>
#   filter(type=='X')
#
# write_csv(bip_2015, 'statcast_2015_bip.csv')
```

Filter the data for balls in play (already did; originally 653k rows...).

```

statcast_2017_bip <- read_csv("statcast_2017_bip.csv")

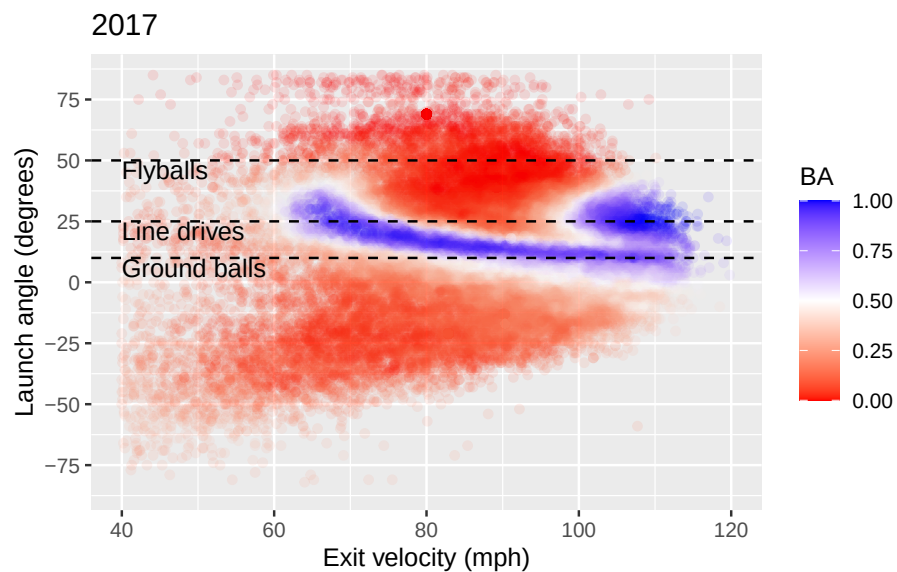
statcast_bip <- statcast_2017_bip |>
  # filter(type == "X") |>
  sample_n(40000) |>
  mutate(p_throws = ifelse(p_throws == "L","P-L","P-R"),
         stand = ifelse(stand == "L","B-L","B-R"))

guidelines <- tibble(
  launch_angle = c(10, 25, 50),
  launch_speed = 40,
  label = c("Ground balls", "Line drives", "Flyballs")
)

ev_plot = statcast_bip |>
  ggplot(aes(x = launch_speed, y = launch_angle,
            color = estimated_ba_using_speedangle)) +
  geom_point(alpha = 0.1) +
  geom_hline(data = guidelines, aes(yintercept = launch_angle),
            color = "black", linetype = 2) +
  geom_text(data = guidelines,
            aes(label = label, y = launch_angle - 4),
            color = "black", hjust = "left") +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=.5) +
  scale_x_continuous("Exit velocity (mph)",
                    limits = c(40, 120)) +
  scale_y_continuous("Launch angle (degrees)",
                    breaks = seq(-75, 75, 25),
                    limits = c(-85, 85)) +
  ggtitle("2017")

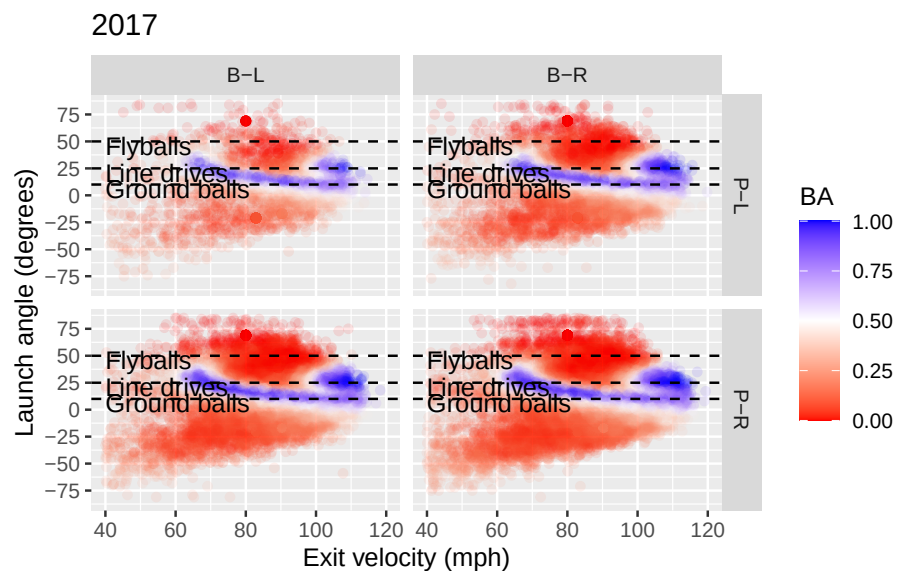
ev_plot

```



Let's break this down by pitcher and batter handedness.

```
ev_plot +  
  facet_grid(p_throws ~ stand)
```



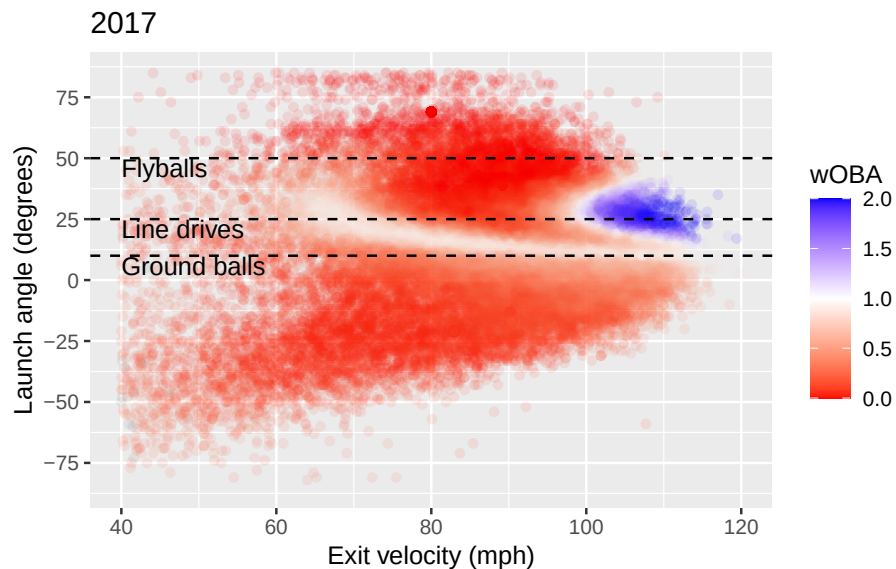
We can also look at velocity and launch angle in terms of wOBA.

From [MLB.com](https://www.mlb.com):

Definition: wOBA is a version of on-base percentage that accounts for how a player reached base – instead of simply considering whether a player reached base. The value for each method of reaching base is determined by how much that event is worth in relation to projected runs scored (example: a double is worth more than a single). For instance: In 2014, a home run was worth 2.101 times on base, while a walk was worth 0.69 times on base. So a player who went 1-for-4 with a home run and a walk would have a wOBA of .558 – $(2.101 + 0.69 / 5 \text{ PAs})$.

The formula: Where “factor” indicates the adjusted run expectancy of a batting event in the context of the season as a whole: (unintentional BB factor x unintentional BB + HBP factor x HBP + 1B factor x 1B + 2B factor x 2B + 3B factor x 3B + HR factor x HR)/(AB + unintentional BB + SF + HBP).

```
ev_plot +
  aes(color = estimated_woba_using_speedangle) +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))
```



Now let's look at the 2015 Data.

```
statcast2015_bip <- read_csv("statcast_2015_bip.csv") |>
  sample_n(40000)
```



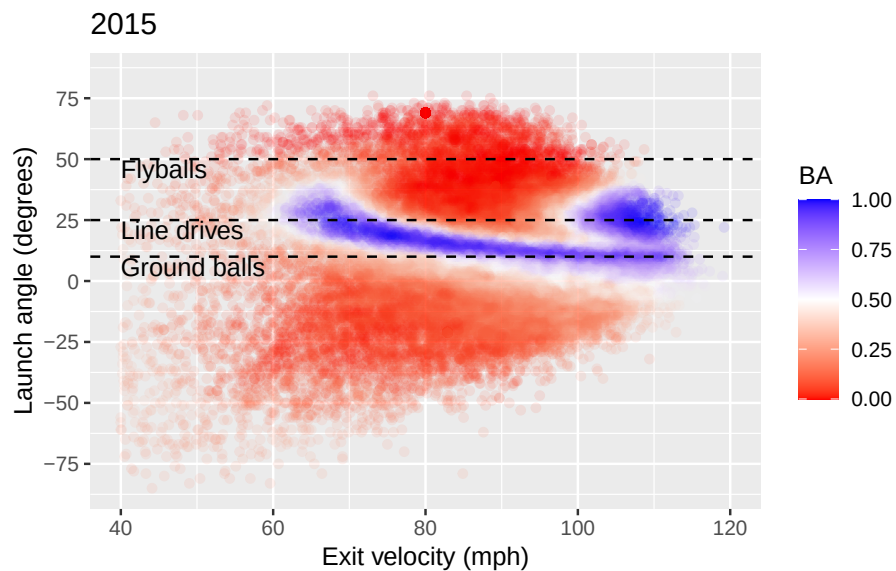
```

ev_plot_2015 = statcast2015_bip |>
  ggplot(aes(x = launch_speed, y = launch_angle,
             color = estimated_ba_using_speedangle)) +
  geom_point(alpha = 0.1) +
  geom_hline(data = guidelines, aes(yintercept = launch_angle),
            color = "black", linetype = 2) +
  geom_text(data = guidelines,
            aes(label = label, y = launch_angle - 4),
            color = "black", hjust = "left") +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=.5) +
  scale_x_continuous("Exit velocity (mph)",
                    limits = c(40, 120)) +
  scale_y_continuous("Launch angle (degrees)",
                    breaks = seq(-75, 75, 25),
                    limits = c(-85, 85)) +

  ggtitle("2015")

ev_plot_2015

```

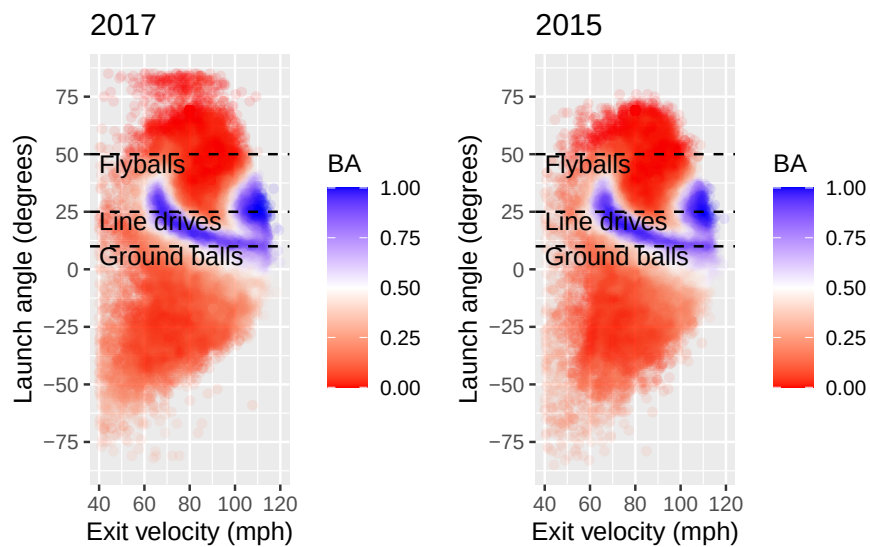


Lets look at these plots together.

```

library(patchwork)
a = ev_plot
b = ev_plot_2015
a+b

```



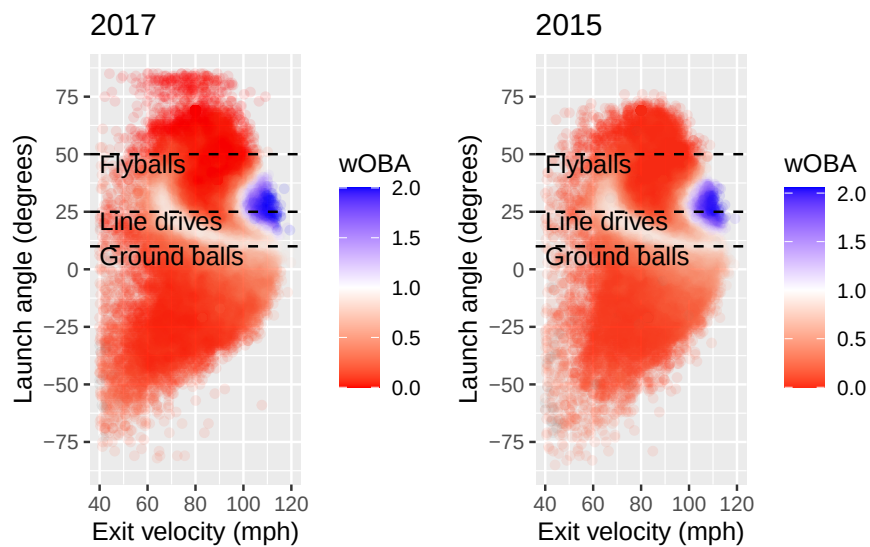
How do these two plots compare?

Let's make the same side-by-side plot for wOBA:

```
c = ev_plot +
  aes(color = estimated_woba_using_speedangle) +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))

d = ev_plot_2015 +
  aes(color = estimated_woba_using_speedangle) +
  scale_color_gradient2("BA", low = "red", mid = "white", high = 'blue',midpoint=1) +
  guides(color = guide_colorbar(title = "wOBA"))

c+d
```



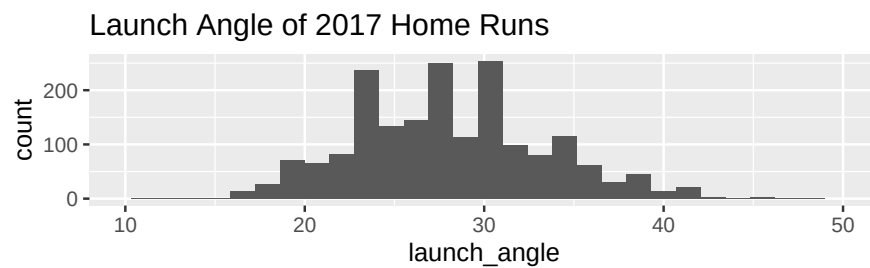
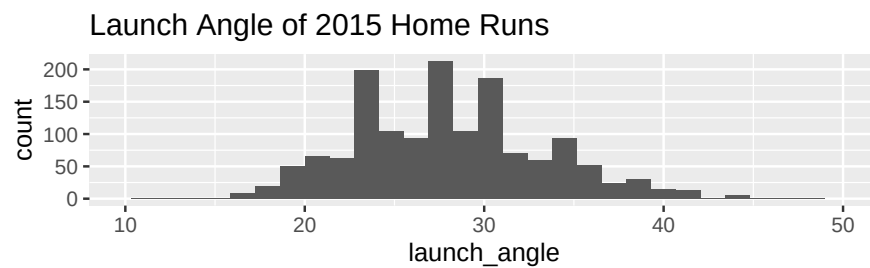
Now let's look at the launch angles and exit velocities for home runs. Construct histograms of these metrics for both 2015 and 2017.

```
hr2017 <- statcast_bip |>
  filter(events == "home_run")
hr2015 <- statcast2015_bip |>
  filter(events == "home_run")

la_2015 <- hr2015 |>
  ggplot(aes(x = launch_angle)) +
    geom_histogram() +
    labs(title = "Launch Angle of 2015 Home Runs") +
    scale_x_continuous(limits = c(10,50))

la_2017 <- hr2017 |>
  ggplot(aes(x = launch_angle)) +
    geom_histogram() +
    labs(title = "Launch Angle of 2017 Home Runs") +
    scale_x_continuous(limits = c(10,50))

la_2015/la_2017
```



```
ev_2015 <- hr2015 |>
  ggplot(aes(x = launch_speed)) +
  geom_histogram() +
  labs(title = "Exit Velocity of 2015 Home Runs",
       y = "Count",
       x = "Exit Velocity") +
  xlim(80,120)

ev_2017 <- hr2017 |>
  ggplot(aes(x = launch_speed)) +
  geom_histogram() +
  labs(title = "Exit Velocity of 2017 Home Runs",
       y = "Count",
       x = "Exit Velocity") +
  xlim(80,120)

ev_2015/ev_2017
```

